Date:_____

Task 1 : Character and ASCII Values

Program 1 :

```
#include <iostream>
using namespace std;
int main() {
char letter;
 letter = 'A';
cout<< letter<< "\n";
letter = 'B';
Cout<< letter<< '\n";
 cin >> letter;
  return 0;
}
```

outBut

A

B

Program 2 :

```
#include<iostream>
using namespace std;
int main() {
 char letter;
 letter = 65;
cout<< letter<<"\n";
 letter = 66;
cout<< letter<< "\n";
 cin>> letter;
  return 0;
}
```

outP

65

6

Program 3:

```
#include <iostream>
using namespace std;
int main() {
    int letter;
    letter = 'A';
Cout<< letter<< "\n";
 letter = 'B';
Cout<< letter<< "\n";
 Cin >> letter;
    return 0;
}
```

outPut

A
B

Task 2: Arithmetic Operation

~~#include <iostream~~

A C++ Program that takes two integers an input and applies addition, subtration, multiplication, division, and modulus.

```
#include <iostream>
using namespace std;
    int main()  {
        int num1, num2;
    Cout<< "Enter two integers:";
    Cin>>num1 >>num2;
 Cout<<"Addition- "<<num1 + num2<<"\n";
 Cout <<" Subtraction-"<<num1 - num2<<"\n";
 Cout<< " Multiplication= "<<num1 * num2<<"\n";
 Cout<<" Division = "<<num1 / num2<<"\n";
```

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```
Cout << "Modulus = " num 1 % num2 << "\n";
    return 0;
}
```

Sample input : 10 5

Sample output:

Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2
Modulus = 0

Task 3: write the output For the following program, if the user enters 10, 20 and 30:

```
#include <iostream>
using namespace std;
    int main() {
    int val 1;
    int val 2;
    int val 3;
Cout<< "Please enter Your 3 numbers.";
Cin >> val 1 >> val 2 >> val 3;
Cout << val1 << "\n" << val2 << "\n" << val3 <<
Cout << val3 << "\n" << val2 << "\n" << val1 << "\n
    system ("PAUSE");
        return 0;
    }
```

OutPut

```
Please enter Your 3 numbers:
10
20
30
30
20
10
```

ack 4:

Distance between two Points

C++ Program to Calculate the distance between two Points using the formula:

$$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$

```
#include <iostream>
#include <cmath>
using namespace std;
int main () {
    double x1, y1, x2, y2, d;
Cout<< "Enter X1 and Y1 : ";
Cin >> x1 >> y1;
Cout << "Enter X2 and Y2 : ";
   Cin>> x2 >> y2;
   d = Sqrt((x2 - x1)* (x2-x1) +
            (y2 - y1)*(y2-y1));
Cout<< "Distance=" << d << endl;
      return 0;
```

Sample input:

0 0

3 4

Sample output

Distance = 5

Task 5:

- Area of a Circle

Formula:

$$Area = PI * r^2$$

Program:

```
#include <iostream>
using namespace std;
int main() {
double radius, area;
Const double PI = 3.14159;
Cout << "Enter radius: ";
Cin >> radius;
area = PI * radius * radius
Cout << "Area of circle = " << area;
return 0;
}
```

output

```
Enter radius: 5
Area of circle = 78.5397
```

- Area of a Rectangle

Formula:

Area = Length x width

Program:

```
#include <iostream>
using namespace std;
int main() {
double length, width, area;
Cout << "Enter length:";
cin >> length;
Cout << "Enter width:";
cin >> width;
area = length * width;
Cout << "Area of Rectangle =" << area;
return 0;
}
```

Output

```
Enter length: 10
Enter width: 5
Area of Rectangle: 50
```

sk 6:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
double item1 = 12.95;
double item2 = 24.95;
double item3 = 6.95;
double item4 = 14.95;
double item5 = 3.95;
double subtol, sales Tex, total;
Subtotal = item1 + item2 + item3 + item4 + item5;
```

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```
item    SalesTax = Subtotal * 0.06;
        Total = Subtotal + SalesTax;
        Cout << Fixed << setPrecision(2);
        Cout << "Price of item 1 = $" << item1 << endl;
        Cout << "Price of item 2 = $" << item2 << endl;
        Cout << "Price of item 3 = $" << item3 << endl;
        Cout << "Price of item 4 = $" << item4 << endl;
        Cout << "Price of item 5 = $" << item5 << endl;
        Cout << "Subtotal = $" << subtotal << endl;
        Cout << "Sales Tax = $" << salesTax << endl;
        Cout << "Total = $" << Total << endl;
            return 0;
        }
```

output

```
Price of item1 - $12.95
Price of item2 = $24.95
Price of item3 - $ 6.95
Price of item4 - $14.95
Price of item5 = $3.95
    subtotal = $63.95
    Sales Tax = $3.83
    Total - $67.58
```

Bonus Task: Break number

Q- A Program that inputs a five-digit integer and prints its individual digits separated by three spaces.

```
#include <iostream>
using namespace std;
int main() {
    int number;
Cout<<"Enter a five-digit number.";
Cin >> number;
Cout<< number /10000<<"\t";
number = number % 10000;
Cout<< number / 1000<<"\t";
number - number % 1000;
Cout<<number / 100<<"\t";
number = number % 100;
Cout<< number / 10 <<"\t";
number = number % 10;
Cout<< number;
return 0;
}
```

outPut

```
Entera five-digit
number: 12345
1 2 3 4 5
```

Home Task 1:

Q- A Program that reads two variables and swaps their values.

```
#include <iostream>
using namespace std;
int main() {
```

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```
int var 1 , var2 , temp;
cout<< "Enter value of var 1.";
 cin>> var1;
Cout<<"Enter value of var 2.";
 cin>> var2;
  temp = var 1;
   var1 = var 2;
   var2 = temp;
  Cout<<"After swapping:"<<endl;
 Cout<<"val1 ="<<Var 1 << endl;
 Cout<<"var2 ="<< var2 << endl;
    return 0;
  }
```

output

```
Enter value of var 1 : 10
Enter value of var 2 : 30
Var 1 : 30
Var 2 : 10
```

Task 2:

ake upper case usin ASCII.

```
#include<iostream>
using namespace std;
  int main() {
  char ch;
 cout<<"Enter a small letter:";
  cin>> ch;
   ch = ch - 32;
   cout<<"Uppercase letter =" <<ch;
    return 0;
  }
```

output

```
Enter a small letter: a
Uppercase letter: A
```

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